

Independent Power Producers (IPPs) / Generators		Gather Input & Prepare Collect plant, fuel, and price data		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Maximize real-time revenue from generation assets Ensure units are available and operating within constraints Manage exposure to real-time price and imbalance risk	Data aggregation
		Actions	Collect real-time data (plant status, outages, generation availability) Monitor market prices (LMP) and dispatch signals from ISO Track actual vs scheduled generation and deviations	
		Thoughts	“Are my units running optimally right now?” “Am I capturing the best price opportunities?” “Do I have any imbalance or penalty risk?”	
		Pain Point	Sudden outages or performance limitations Volatile real-time prices Limited control over dispatch decisions (ISO-driven)	
		Emotions	Alert to capture price spikes Stress during outages or price drops Satisfaction when assets perform profitably	
		Tool	SCADA / plant monitoring systems Real-time market dashboards (LMP, dispatch) PLEXOS / dispatch tracking tools	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Optimize bidding strategy to maximize revenue Forecast prices and position assets competitively Plan asset utilization and investment decisions	Example AI Capability AI combines plant + market + fuel data into one view
		Actions	Gather inputs (fuel cost, plant efficiency, outage schedules, market forecasts) Analyse historical and forecasted price trends Prepare bids/offers for DAM based on expected profitability	
		Thoughts	“What price should I bid to maximize profit?” “How will market conditions impact my revenue?” “Are my assets competitive in this market?”	
		Pain Point	Uncertainty in price forecasts and market behaviour Competition from other generators and renewables Fuel price volatility impacting margins	
		Emotions	Analytical when preparing bids Uncertain about market outcomes Confident when strategy is well-aligned with forecasts	
		Tool	PLEXOS (bidding strategy & price forecasting) Market data platforms (prices, demand, competitors) Fuel and cost modelling tools	

Independent Power Producers (IPPs) / Generators		Simulation Setting Configure bidding & asset models		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Configure short-term model to optimize real-time dispatch and revenue Reflect current plant availability, constraints, and market conditions Generate outputs to track performance and imbalance risk	Strategy suggestion
		Actions	Set short-term horizon (intra-hour / hourly) and resolution Input real-time data (unit availability, outages, fuel limits, dispatch signals) Select outputs (price, dispatch, generation, imbalance, revenue impact)	
		Thoughts	“Is the model capturing current plant constraints correctly?” “Will outputs reflect actual dispatch and price signals?” “Am I capturing revenue and risk accurately?”	
		Pain Point	Incorrect assumptions lead to wrong dispatch signals Limited visibility into market-wide conditions Too many outputs make it hard to focus on key revenue drivers	
		Emotions	Careful while configuring high-impact runs Pressure to align model with real-time conditions Confidence when setup reflects actual operations	
		Tool	PLEXOS short-term dispatch / simulation SCADA / plant monitoring systems Real-time market dashboards (LMP, dispatch signals)	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Configure model for optimal bidding strategy and revenue maximization Reflect asset characteristics, fuel costs, and market conditions Generate outputs for price forecast, dispatch, and profitability	Example AI Capability AI suggests bid curves based on cost + market trends
		Actions	Define horizon (day-ahead, week-ahead) and scenario assumptions Input bids/offers, fuel costs, plant constraints, and market forecasts Select outputs (price forecast, unit commitment, revenue, utilization)	
		Thoughts	“Are my cost assumptions and bids realistic?” “Will this model reflect competitive market behaviour?” “Can I rely on this for bidding decisions?”	
		Pain Point	Complexity in modelling bidding strategies and market response Uncertainty in price forecasts and competitor behaviour Difficulty aligning model with actual market outcomes	
		Emotions	Analytical while setting up model Uncertain about market assumptions Confident when setup supports clear bidding strategy	
		Tool	PLEXOS (bidding strategy & market simulation) Market data platforms (prices, demand, competitors) Fuel cost and asset performance models	

Independent Power Producers (IPPs) / Generators		Execute Back cast Model & RUN Validate price, dispatch & revenue vs history		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Validate how assets performed under past real-time market conditions Check dispatch accuracy and imbalance exposure Improve confidence in short-term operational strategies	Revenue gap analysis
		Actions	Load historical data (prices, dispatch, outages, generation output) Run back cast for selected periods (peak hours, price spikes) Compare simulated vs actual generation, dispatch, and revenue	
		Thoughts	“Did my assets perform as expected historically?” “Where did I lose revenue or face imbalance penalties?” “Can I trust this model for real-time decisions?”	
		Pain Point	Incomplete or inconsistent historical plant data Difficulty capturing real-time price volatility Hard to isolate performance issues (asset vs market vs model)	
		Emotions	Focused during performance validation Frustrated when results deviate from actual outcomes Confident when model aligns with historical behaviour	
		Tool	PLEXOS back casting / dispatch simulation Historical plant and market data (prices, dispatch) SCADA / plant performance data	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Validate price forecasts and bidding strategies against historical outcomes Benchmark revenue and asset utilization performance Build a reliable base model for future bidding and planning	Example AI Capability “You missed 12% revenue during peak hours”
		Actions	Run back cast using historical prices, demand, fuel cost, and constraints Compare simulated vs actual DAM results (price, dispatch, revenue) Adjust assumptions and rerun to improve alignment	
		Thoughts	“Is my price forecast close to actual market outcomes?” “Did my bidding strategy capture available revenue?” “Is this model reliable for future decisions?”	
		Pain Point	Difficulty capturing competitor behaviour and market dynamics Gaps between modelled and actual price outcomes Need high accuracy for revenue forecasting	
		Emotions	Analytical while benchmarking performance Uncertain when results diverge Confident when model reflects real market behaviour	
		Tool	PLEXOS market simulation / back casting Historical market data (prices, demand, bids if available) Excel / analytics tools for revenue comparison	

Independent Power Producers (IPPs) / Generators		Analyse Solution Analyse profit drivers and asset performance		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time	Goal	Evaluate real-time asset performance and revenue outcomes Identify imbalance exposure and dispatch efficiency Ensure assets are operating at optimal levels	Profit insights
		Real-time dispatch & generation report		
		Price (LMP) tracking & opportunity report	Actions Analyse outputs (generation, dispatch, LMP, imbalance, revenue) Compare expected vs actual plant performance Identify inefficiencies or missed revenue opportunities	
		Imbalance & deviation report	Thoughts “Am I generating at the most profitable times?” “Did I miss any high-price opportunities?” “Is my plant operating efficiently?”	
		Plant performance / availability report		
		Revenue realization (hourly) report	Pain Point Limited control over dispatch (ISO-driven decisions) Difficulty linking price signals to asset performance Rapid price fluctuations affecting revenue	
			Emotions Alert when analysing price and dispatch signals Frustration when revenue opportunities are missed Satisfaction when assets perform optimally	
			Tool PLEXOS output analysis tools Real-time market dashboards (LMP, dispatch) SCADA / plant performance monitoring systems	
	Day Ahead / Forward	Goal	Evaluate profitability of bidding strategies and scenarios Understand key drivers of revenue (price, cost, utilization) Support decisions on asset utilization and investment	Example AI Capability AI explains: “Low utilization due to high marginal cost”
		Price forecast report (DA / forward curves)		
		Bidding strategy & offer curve report	Actions Analyse outputs (price forecast, unit commitment, revenue, utilization) Compare performance across different scenarios Identify drivers of profit and risk exposure	
		Unit commitment & dispatch forecast	Thoughts “Which scenario maximizes my revenue?” “What factors are impacting my margins?” “Are my assets competitive in this market?”	
		Revenue & margin forecast report		
		Risk exposure (VaR / downside scenarios)	Pain Point Complexity in interpreting price vs cost vs dispatch Uncertainty in future market conditions Difficulty benchmarking against competitors	
		Portfolio optimization report	Emotions Analytical while evaluating results Uncertain when results vary across scenarios Confident when clear profit drivers are identified	
			Tool PLEXOS output & scenario analysis Excel / BI tools for revenue and cost analysis Market data platforms (prices, demand, competitors)	

Independent Power Producers (IPPs) / Generators		Forecast/Base Model & Run Forecast prices and revenue opportunities		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Predict short-term price movements and dispatch opportunities Anticipate imbalance risks and optimize real-time generation Maximize revenue from near-term market conditions	Price forecasting
		Actions	Run short-term forecasts using latest price, load, and outage data Update base model with current plant constraints and system conditions Generate outputs (price forecast, dispatch, imbalance exposure, revenue)	
		Thoughts	“Will prices increase in the next few hours?” “Should I adjust generation to capture higher revenue?” “Am I exposed to imbalance penalties?”	
		Pain Point	High volatility in real-time prices Limited control over dispatch decisions (ISO-driven) Forecast errors impacting revenue opportunities	
		Emotions	Alert to capture price spikes Uncertain due to market volatility Confident when forecasts align with trends	
		Tool	PLEXOS short-term forecasting / dispatch models Real-time market data feeds (LMP, demand) SCADA / plant monitoring systems	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Forecast prices and market conditions to maximize bidding revenue Position assets competitively in the day-ahead market Support investment, hedging, and portfolio planning decisions	Example AI Capability AI predicts LMP curves for next day/week
		Actions	Run base model using forecasted demand, fuel cost, outages, and constraints Generate price forecasts, unit commitment, and revenue projections Evaluate multiple assumptions (demand, renewables, fuel, competition)	
		Thoughts	“What price should I bid tomorrow?” “How will market conditions affect my revenue?” “Are my assets competitive under these forecasts?”	
		Pain Point	Uncertainty in price forecasts and competitor behaviour Fuel cost variability impacting margins Difficulty predicting market clearing outcomes	
		Emotions	Analytical during forecasting Uncertain due to market unpredictability Confident when forecasts support strong strategy	
		Tool	PLEXOS (price forecasting & market simulation) Market data platforms (prices, demand, competitors) Fuel cost and asset performance models	

Independent Power Producers (IPPs) / Generators		Forecast/Base Model + Scenario-A,,, & Run Test profit vs risk across scenarios		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Evaluate short-term revenue opportunities under different market conditions Identify risk scenarios (low prices, outages, imbalance penalties) Prepare optimal real-time response strategies	Profit-risk simulation
		Actions	Create scenarios (price spike, demand surge, outage, renewable drop) Run simulations with updated plant constraints and system conditions Compare outputs (price, dispatch, revenue, imbalance exposure)	
		Thoughts	“Which scenario gives me the highest revenue?” “What is my downside risk if prices fall?” “Should I adjust generation strategy now?”	
		Pain Point	High volatility makes scenario outcomes unpredictable Limited control over dispatch decisions Difficulty identifying most critical scenarios quickly	
		Emotions	Alert while evaluating revenue opportunities Pressure due to rapid market changes Confidence when optimal strategy is identified	
		Tool	PLEXOS short-term scenario simulation Real-time market dashboards (LMP, dispatch) SCADA / plant monitoring systems	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Identify optimal bidding strategy across multiple scenarios Understand revenue sensitivity to market conditions Reduce risk while maximizing expected profit	Example AI Capability AI shows revenue under different price/fuel scenarios
		Actions	Define scenarios (high/low demand, fuel cost changes, renewable variability, competition) Run simulations to generate price, dispatch, and revenue outcomes Evaluate results to identify best bidding and hedging strategy	
		Thoughts	“Which scenario gives the best risk-adjusted return?” “How sensitive is my revenue to price and fuel changes?” “What is the safest yet profitable bidding strategy?”	
		Pain Point	Managing and interpreting multiple scenarios Uncertainty in competitor behaviour and market dynamics Difficulty balancing risk vs reward	
		Emotions	Analytical during scenario evaluation Uncertain when results vary widely Confident when clear strategy emerges	
		Tool	PLEXOS scenario modelling & simulation Market data platforms (prices, demand, competitors) Fuel cost and financial risk modelling tools	

Independent Power Producers (IPPs) / Generators		Compare Scenarios Select highest risk adjusted return strategy		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Identify scenario with highest short-term revenue potential Evaluate downside risk (low prices, imbalance penalties) Select best operational strategy for immediate execution	Strategy recommendation
		Actions	Compare outputs (price, dispatch, generation, revenue, imbalance) Identify best-case vs worst-case revenue scenarios Select optimal generation and dispatch approach	
		Thoughts	“Which scenario gives me the most profit right now?” “What is my downside risk if prices fall?” “Should I change generation strategy immediately?”	
		Pain Point	High volatility makes comparison difficult Limited control over final dispatch (ISO-driven) Small price differences can significantly impact revenue	
		Emotions	Alert while evaluating opportunities Pressure to react quickly to market signals Confidence when best revenue scenario is clear	
		Tool	PLEXOS scenario comparison outputs Real-time market dashboards (LMP, dispatch) SCADA / plant performance systems	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Identify optimal bidding strategy across multiple scenarios Balance expected profit with risk exposure Support investment and hedging decisions	Example AI Capability AI suggests: “Bid higher during peak hours for +8% revenue”
		Actions	Compare outputs (price forecast, unit commitment, revenue, utilization) Analyse sensitivity across demand, fuel, renewable, and competition scenarios Select best strategy for DAM bidding and portfolio positioning	
		Thoughts	“Which scenario gives the best risk-adjusted return?” “How sensitive is my revenue to key assumptions?” “Which strategy is safest yet profitable?”	
		Pain Point	Managing multiple scenarios and interpreting results Uncertainty in competitor behaviour and market outcomes Difficulty balancing profit vs risk	
		Emotions	Analytical during comparison Uncertain when scenarios conflict Confidence when optimal strategy emerges	
		Tool	PLEXOS scenario comparison & reporting Excel / BI dashboards for revenue comparison Market data platforms (prices, demand, competitors)	

Independent Power Producers (IPPs) / Generators		Results Execute bids and realize revenue		Where AI Helps
Independent Power Producers (IPPs) / Generators (Focused on asset value, dispatch, revenue, and market risk) Operator Trader Market Analyst Portfolio Manager Risk Manager Planner Investment Analyst	Real Time Real-time dispatch & generation report Price (LMP) tracking & opportunity report Imbalance & deviation report Plant performance / availability report Revenue realization (hourly) report	Goal	Maximize realized revenue from real-time operations Ensure assets operate within limits while minimizing imbalance penalties Respond effectively to real-time market conditions	Performance tracking
		Actions	Monitor actual dispatch, generation, price, and imbalance outcomes Adjust generation based on real-time price signals and system conditions Track revenue performance and penalty exposure	
		Thoughts	“Did I capture the best available price?” “Am I incurring any imbalance penalties?” “Do I need to adjust operations immediately?”	
		Pain Point	Limited control over dispatch decisions (ISO-driven) Price volatility impacting realized revenue Difficulty linking outcomes to prior decisions	
		Emotions	Satisfaction when high prices are captured Stress during low-price or high-penalty periods Confidence when operations perform as expected	
		Tool	SCADA / plant monitoring systems Real-time market dashboards (LMP, dispatch) Performance tracking and revenue dashboards	
	Day Ahead / Forward Price forecast report (DA / forward curves) Bidding strategy & offer curve report Unit commitment & dispatch forecast Revenue & margin forecast report Risk exposure (VaR / downside scenarios) Portfolio optimization report	Goal	Deliver final bidding outcomes and maximize revenue realization Evaluate profitability of strategy and improve future decisions Support long-term investment and hedging strategies	Example AI Capability AI tracks real-time profit vs forecast
		Actions	Submit bids/offers and review market clearing results (price, dispatch) Analyse revenue, utilization, and asset performance outcomes Feed results into future strategy, forecasting, and portfolio planning	
		Thoughts	“Did my bidding strategy maximize profit?” “How did actual results compare to forecasts?” “What should I improve for the next cycle?”	
		Pain Point	Gap between forecasted and actual market outcomes Competitive market dynamics reducing expected margins Need to justify performance internally	
		Emotions	Satisfaction when strategy performs well Pressure from revenue expectations Confidence when results are consistent and explainable	
		Tool	PLEXOS output and reporting tools Market Management Systems (DAM results) BI / analytics dashboards for revenue and performance	